ECON 0100 | HW D1 | Game Theory

Due: Sunday, Oct. 25

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | Dorm Room Kitchen

Harry and Ron share a dorm at Hogwarts and like to keep their kitchen stocked with snacks. They each have the option to contribute or not contribute to the kitchen snacks every week. They both like having the kitchen stocked even if they don't eat the snacks but neither likes the chore of buying the snacks. If they both contribute, the kitchen will be fully stocked and they both receive a payoff equivalent to 6 galleons. If only one of them contributes, the contributor is able to enjoy the partially stocked kitchen but also had to do the shopping, receiving a payoff equivalent to 2 galleons, while the other receives a payoff equivalent to 10 galleons. If neither of them contributes, the kitchen is empty and they both receive a payoff of 0 galleons.

		Ron	
		Contribute	Don't Contribute
Harry	Contribute	6, 6	2, 10
	Don't Contribute	10, 2	0, 0

a) Identify Ron's best response.

b) Identify Ron's dominant strategy if one exists.

c) Identify a Nash Equilibrium in this game if one exists.

d) Find the socially efficient strategies.

Q2 | Late-Night Wireless

Seamus and Dean share a dorm at Hogwarts. Each night, each chooses whether to play his wireless late or keep quiet so the other can sleep. Each enjoys his own music, but both value a good night's sleep, and they don't value them equally. Their payoffs, in galleons, are below.

		Dean	
		Quiet	Play
Seamus	Quiet	6, 5	1, 7
	Play	8, 2	4, 3

a) Identify Dean's best response.

b) Identify Dean's dominant strategy if one exists.

c) Identify a Nash Equilibrium in this game if one exists.

d) Find the socially efficient strategies.

Q3 | Sunday Breakfast

Fred and George like a big breakfast at the Burrow on Sunday mornings, but it takes two cooks. Each twin chooses whether to get up early to cook or to sleep in and make toast later. Their payoffs, in galleons, are below.

		George	
		Cook	Sleep In
Fred	Cook	8, 8	0, 4
	Sleep In	4, 0	4, 4

a) Identify George's best response.

b) Identify George's dominant strategy if one exists.

c) Identify a Nash Equilibrium in this game if one exists.

d) Find the socially efficient strategies.